MILITARY SPECIFICATION

MODULE, PLUG-IN, AUTOMATIC LEAD SELECTOR-IN-CIRCUIT CHECKER FOR TEST SET, TRANSISTOR: AN/USM - 136

1. SCOPE

1.1 This specification covers the Automatic Lead Selector - In-Circuit Checker plug-in module used in conjunction with the Transistor Test Set, AN/USM-136 which is one type of transistor test set used to test transistors, semiconductor diodes and other semiconductor devices. The plug-in Automatic Lead Selector - In-Circuit Checker shall be hereinafter referred to as the "module".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-945	Test Equipment for Use with Electronic Equipment; General Specification for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-14252	Fire Control Materiel: General Specification Governing the Design and Inspection of
MIL-I-45208	Inspection Requirements: General Specification for
MIL-T-55202	Test Set, Transistor AN/USM-136

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

A8625349	Assembly, Test Set, Transistor; AN/USM-136
F8625449	Automatic Lead Selector - In-Circuit Checker
C8625532	Cross Reference, Electrical Schematics

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. - Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication. - The module shall be manufactured in accordance with Drawing F8625449 "Assembly, Electrical Equipment, Automatic Lead Selector" and all drawings and specifications pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence to insure inherent quality and performance capabilities.

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3.4 General specifications. - The contractor shall be responsible for adherence to, and compliance with the following specification: MIL-T-945, MIL-E-5272, MIL-F-13926, and MIL-F-14252, particularly the following paragraphs:

3.4.1 Specification MIL-E-5272

(a) High temperature test, Procedure II

(b) Low temperature test, Procedures I and II

(c) Temperature shock test, Procedure I

(d) Humidity test, Procedure I

(e) Altitude test, Procedure VI

(f) Shock, Procedure V, equipment operation

3.4.2 Specification MIL-F-13926

(a) Order of precedence

(b) Protective finished surfaces

(c) Lubrication

3.4.3 Specification MIL-F-14252

(a) Use of standard items

(b) Precedence of standard drawings

(c) Wired equipment

(d) Workmanship

3.5 Environmental.

3.5.1 Operating temperatures. - The module shall meet all the performance requirements of this specification, when thermally stabilized at ambient temperatures of -20°C (-4°F) to +50°C (+122°F).

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3.5.2 Storage temperatures. - The module shall meet all the performance requirements of this specification upon being thermally stabilized at operating temperature after having been thermally stabilized at ambient temperatures of -62°C (-80°F) and +71°C (+160°F).

3.5.3 Operating altitude. - The module shall meet all the performance requirements of this specification, when thermally stabilized at a temperature of -20°C (-40°F) and a pressure of 20.58" Hg. (10,000 ft. altitude).

3.5.4 Non-operating altitude. - The module shall meet all the performance requirements of this specification upon return to room temperature of +25°C (+77°F) after being subjected to a temperature shock test, the limits of which are -40°C (-40°F) and +85°C (+185°F).

3.5.5 Temperature shock. - The module shall be capable of meeting all the performance requirements of this specification upon return to room temperature of +25°C (+77°F) after being subjected to a temperature shock test, the limits of which are - 40°C (-40°F) and +85°C (+185°F).

3.5.6 Shock. - The module shall be capable of withstanding 18 shocks of 15 G's acceleration for a duration of 11 plus or minus 1 millisecond. The module shall not suffer damage or subsequently fail to provide the performance specified in this specification.

3.5.7 Vibration. - The module shall be capable of meeting the performance requirements of this specification after being subjected to the following vibration tests.

(a) A simple harmonic motion of 0.06 inch at a frequency varying from 10 to 33 cycles per second and back to 10 cycles per second.

(b) A simple harmonic motion of 0.03 inch at a frequency varying from 10 to 55 cycles per second and back to 10 cycles per second.

(c) Vibration at the four most severe resonant frequency points.

3.5.8 Humidity. - The module shall meet all the performance requirements of this specification after being subjected to ten cycles of the humidity test described in paragraph 4.9.1.1.8 of this specification.

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3.6 Performance requirements.

3.6.1 Circuitry. - The module circuitry shall be in accordance with the electrical schematic specified on Drawing C8625532, "Cross Reference". The module should be capable of performing the following lead finding tests on all presently known three terminal transistors and two terminal diodes that are specified within the performance envelope limits of this module on an "out-of-circuit" basis. In addition, the module should have the capability of checking the same transistors and diodes "in-circuit" where feasible.

3.6.2 Warm-up. - The module shall be fully operable within 2 minutes after the main power is turned on.

3.6.3 Silicon - germanium finding. - The first function required of the module shall be a determination of Silicon vs. Germanium that shall make an adjustment of the test voltage for the Base finding function which follows: A forward biased Silicon junction shall be found as "Silicon" and cause the base finding test voltage to be between 1.1 volts and 1.7 volts. A forward biased Germanium junction shall be found as "Germanium" and cause the base finding test voltage to be between 0.5 volts and 0.8 volts.

3.6.4 Base (Transistor) or cathode (Diode) finding. - The second function required of the module shall be a determination of which lead is the Base junction of a three terminal transistor of either silicon or germanium, PNP or NPN. It shall determine which lead is the Cathode junction in the case of a diode of either silicon or germanium. The determination is based on the apparent resistance of the junction where a reversed biased junction shunted with resistance no less than 9100 ohms for either germanium or silicon shall be found is a "Base" or "Cathode". An additional requirement of this base finder for in-circuit test applications makes use of the IN-CKT IMP (in-circuit impedance) switch which, when placed in the LOW position, shall reduce the sensitivity and allow the base (cathode) to be found when the shunting resistance is reduced to no less than 250 ohms for germanium or no less than 500 ohms for silicon. At lower than the above values of shunt resistance, if a base (cathode) cannot be found, then a SHORT OR LOW IMP (low impedance) shall be found and so indicated by a light.

3.6.4.1 An additional requirements of the Base finding circuit is a test for OPEN circuits by the use of a relay logic and indicators, the Base (cathode) finding circuit shall determine if an open circuit exists between any of the three junctions of a transistor or the junction of a diode.

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3.6.5 Emitter finding. - The third function required of the module shall be an out-of-circuit determination of which lead is the Emitter of a three terminal transistor having a forward gain (Beta) larger than its reverse gain (Inverse Beta) and of either silicon or germanium, PNP or NPN.

3.6.5.1 An additional requirement of the emitter finding circuit is a NO-GAIN check. By the use of a relay logic and indicator, the Emitter finding circuit shall indicate a "NO-GAIN" condition if either; (1) no gain is found or (2) a short circuit or low impedance exists across the emitter to collector junction of the transistor preventing it from being a gainful device or (3) if the forward gain and reverse gain appear to be approximately equal. An additional requirement, for in-circuit test applications with shunt impedances lower than 9,100 ohms but not below 250 ohms for germanium or 500 ohms for silicon between any two leads; the IN-CIRCUIT switch, when closed, shall alter the circuit from one of emitter finding to one of forward vs. reverse gain difference detection. In this configuration the Emitter finding circuit is less affected by shunts and shall also indicate "NO-GAIN" for the above three reasons, however, it is not used for the correct determination of the emitter lead.

3.6.6 PNP-NPN finding. - The fourth function required of the module shall be a determination of PNP polarity or NPN polarity of a good three terminal transistor of either silicon or germanium on an in-circuit basis where the shunt resistances is not less than 250 ohms for germanium or 500 ohms for silicon. The PNP or NPN polarity is correctly programmed into the module, when used in the Transistor Test Set AN/USM-136 with punched card programming for out-of-circuit testing of transistors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. - Unless otherwise specified the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the Contractor order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. - Inspections shall be in accordance with Specification MIL-I-45208.

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4.3 Initial production sample. - Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the Contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Initial production sample failure. - Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. - Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots. - Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material. - Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components. - All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of transistor test set shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as requirements effecting concentricity, squareness, parallelism, microfinishes 64 RMS or better, threads, identification marking and performance.

4.6.1 Sampling inspection. - Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table I.

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TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.6.2 Disposition of non-conforming product. - Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.7 Inspection equipment.

4.7.1 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes established by product requirement shall be capable of measuring those values within an accuracy of ten percent of specified tolerance, except as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 Test equipment accuracy. - Test equipment utilized in this specification shall contain the following accuracies:

Item	Description	Accuracy Requirements
1.	DC Power Supply	Negative 30 volt plus or minus 10 percent, current rating greater than or equal to 2 amperes, peak-to-peak ripple less than 2 volts, output impedance less than 2 ohms from DC to 2 kilocycles.
2.	AC Power Supply	2.5 volts RMS, plus or minus 3 percent, current rating greater than or equal to 50 milliamperes, 60 cycles per second plus or minus 3 cycles per second.

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Item	Description	Accuracy Requirements		
3.	DC Voltmeter	1.7 volt range	±3%	
4.	Test Resistors	9100 ohm	±5%	1/2 watt
		500 ohm	±1%	1/2 watt
		250 ohm	±1%	1/2 watt
		47 ohm	±5%	1/2 watt

4.7.3 Inspection equipment design requirements. - The accuracy requirements for test equipment to perform environmental tests are specified in MIL-E-5272.

4.8 Acceptance inspection.

4.8.1 Classification of defects. - The classification of defects in Table II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect any applicable requirement, and to reject individual non-conforming items. The AQL values specified pertain to each individual characteristic.

4.8.2 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government informed of the corrective action taken, the lot shall be resubmitted for acceptance, as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II in Table II-A with Sampling Plan Table IV-A of Standard MIL-STD-105.

Critical: None defined.

Major: AQL 1.0%		Req't Par.	Q.A. Par.	Test Methods
101.	Completeness of assembly	3.5	--	Visual
102.	Warm-up	3.6.2	4.9.2.1	Standard Measuring Equipment (SME)
103.	Silicon-Germanium finding	3.6.3	4.9.2.2.1	SME
104.	Base or cathode finding	3.6.4	4.9.2.2.2	SME
105.	Emitter finding	3.6.5	4.9.2.2.3	SME
106.	PNP-NPN finding	3.6.6	4.9.2.2.4	SME

Minor: None defined.

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4.9 Acceptance inspection testing.

4.9.1 Environmental testing. - Environmental tests outlined below shall be performed on the initial production sample and one in every fifty; thereafter, the quantities specified above shall prevail provided the contractor continues manufacturing the item using the same processes and techniques used to produce the initial production sample. Any alteration in the manufacturing techniques or processes will require a new production sample and the sampling procedure shall begin again. The contractor shall be required to present a new production sample when any contract revision, determined by the contracting officer, indicates a change in performance requirements. When the module is part of a procurement for a complete Transistor Test Set AN/USM-136, the environmental tests will be conducted on the module while installed in the complete test set in accordance with the requirements of Specification MIL-T-55202, and the separate environmental tests herein specified shall not be performed. Where the module is part of a separate or spare parts procurement, then the following environmental tests apply:

4.9.1.1 Low temperature.

4.9.1.1.1 Low temperature procedure I. - The module shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -20°C (-4°F) until stabilization of the module temperature is reached. While at this temperature, the module shall be tested to determine conformance to the requirements of this specification.

4.9.1.1.2 Low temperature procedure II. - Following the test in 4.9.1.1.1, the temperature within the chamber shall be reduced to and maintained at a temperature of -62°C (-80°F) for 72 hours with the module de-energized. The temperature of the chamber shall then be raised slowly to the laboratory ambient temperature and the module shall meet the performance requirements of this specification.

4.9.1.1.3 High temperature. - The module shall be placed within the test chamber and the chamber temperature raised to +71°C (+160°F). The relative humidity within the chamber shall not be more than 15%. The module shall be maintained at this temperature for a period of 48 hours. At the conclusion of the exposure period the temperature within the chamber shall be lowered to +50 C (+122 F). After module stabilization at +50°C (+122°F), the module shall be energized and operated at this temperature and shall meet the performance requirements of this specification. The chamber temperature shall then be lowered to the laboratory ambient temperature and a visual inspection performed to determine if any deterioration has occurred.

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4.9.1.1.4 Altitude. - The module shall be placed in the test chamber and the internal chamber temperature reduced to -54°C (-65°F). The internal pressure of the chamber shall be reduced to 3.44 In. Hg. (50,000 ft. altitude). The module shall be maintained under these conditions for one hour. At the conclusion of this period, the internal chamber temperature shall be raised to -20°C (-4°F) and the internal pressure of the chamber shall be raised to 20.58 In. Hg. (10,000 ft. altitude). After the module has stabilized at the new temperature and pressure, the module shall be energized and shall meet the performance requirements of this specification. The chamber shall then be returned to laboratory ambient and pressure and the module inspected visually for damage.

4.9.1.1.5 Temperature shock. - The module shall first be placed within a test chamber wherein a temperature of 85°C (185°F) is maintained. The module shall be exposed to this temperature for a period of 4 hours, at the conclusion of which and within 5 minutes, the module shall be transferred to a chamber having an internal temperature of -40°C (-40°F). The module shall be subjected to this temperature for a period of 4 hours. This constitutes one cycle. The number of complete cycles shall be three. The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the module shall be removed from the test chamber, returned to standard temperature, and within a period of 1 hour shall be visually inspected and shall meet the performance requirements of this specification.

4.9.1.1.6 Shock. - The module, non-operating, mounted in a test fixture to simulate the way in which it is normally mounted in service, shall be subjected to the shock conditions specified herein. The module shall be subjected to 18 impact shocks of 15g acceleration, each shock impulse having a time duration of 11 plus or minus 1 milliseconds. The maximum acceleration shall be reached in approximately 5-1/2 milliseconds. The shock shall be applied in the following directions:

(a) Vertically, 3 shocks in each direction

(b) Parallel to the major horizontal axis, 3 shocks in each direction

(c) Parallel to the minor horizontal axis, 3 shocks in each direction

The module shall be inspected to assure that it did not suffer damage and that it will provide the performance specified.

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4.9.1.1.7 Vibration. - The module, non-operating, mounted in a test fixture to simulate the way in which it is normally mounted in service, shall be vibrated to the following tests.

4.9.1.1.7.1 Vibration test 1. - The module shall be vibrated with the frequency varying between 10 and 33 cycles per second at a total excursion of 0.06 inch, the frequency varying uniformly from 10 to 33 cycles per second and returning to 10 cycles per second in approximately 1/2 minute.

4.9.1.1.7.2 Vibration test 2. - The module shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03 inch, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately 1 minute. The module shall be vibrated for 15 minutes in each of three mutually perpendicular planes for both Vibration Test 1 and Vibration Test 2.

4.9.1.1.7.3 Vibration test 3. - The module shall be vibrated for a period of 3 minutes at each of the four most severe resonant frequency points at an amplitude of 0.03 inch. The module shall then be inspected and shall meet the performance requirements of this specification to determine mechanical or electrical failures.

4.9.1.1.8 Humidity. - The module shall be placed in the test chamber and set up to simulate installed conditions. During the first 2 hour period, the temperature in the test chamber shall be raised from 28°C (82°F) to 71°C (160°F). The temperature of 71°C (160°F) and a relative humidity of 95 percent shall be maintained in the test chamber during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be reduced at a uniform rate to 28°C (82°F) which constitutes one cycle. The cycle shall be repeated 10 times for a total test time of 240 hours. At the conclusion of the 240 hour period, the module shall be returned to laboratory environment and shall meet the requirements of this specification. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity.

4.9.2 Examination.

4.9.2.1 Test conditions. - Unless otherwise specified, all tests shall be performed under the following conditions:

(a) Ambient Conditions; the tests shall be performed under factory or laboratory ambient conditions.

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(b) Insert the Automatic Lead Selector - In-Circuit Checker module, (A100) into the test circuit of Figure 1 (or test while installed in the Transistor Test Set AN/USM-136). The indicator light layout shown as part of Figure 1 must be used to determine lead finding action.

(c) Input Power; the input power shall be as shown in the test circuit of Figure 1 or if tested in the Transistor Test Set AN/USM-136, 115 volts plus or minus 2 percent 60 cycles per second plus or minus 2 percent input to the Transistor Test Set AN/USM-136.

(d) Warm-up; the module shall be turned ON and permitted to warm up for 2 minutes before any test is conducted.

4.9.2.2 Automatic lead selector - In-circuit checker. - Testing the module shall be done in the following manner:

4.9.2.2.1 Silicon - germanium finding.

(a) Switch the LEADS switch to the UNKNOWN position. With the power turned on, press the RESET button and observe the lighting of the READY light.

(b) Use the following test switch line-up: Switch to NPN; TRANSISTOR; LEADS UNKNOWN; OUT-OF-CIRCUIT; IN-CKT IMP (in-circuit impedance) switch in HIGH position. Insert a known good general purpose silicon NPN transistor such as a 2N338 or equivalent into the front panel TRANSISTOR test socket (J103). Place the probes of a DC Voltmeter between Pin 4 of K103 and the chassis ground. Set the meter to read about 1.7 volts negative.

(c) Press the TEST button and during the third, fourth, and fifth "click" of the stepper, the voltage read on the voltmeter shall be between 1.1 volts and 1.7 volts negative.

(d) RESET the module to READY. Replace the test transistor with one known as a good general purpose germanium PNP transistor such as a 2N404 or equivalent at the same location. Switch to PNP.

(e) Press the TEST button and during the third, fourth and fifth "click" of the stepper, the voltage read on the voltmeter shall be between 0.5 volts and 0.8 volts negative. Remove the voltmeter.

(f) Remove the test transistor and RESET the module to READY.

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4.9.2.2.2 Base (Transistor) or cathode (Diode) finding.

(a) Place a known good general purpose germanium PNP transistor such as a 2N404 or equivalent into the front panel TRANSISTOR test socket (J103) with the base in position X, the emitter in position Y and the collector in position Z. Place a 9100 ohm plus or minus 5 percent test resistor across the front panel DIODE test clips (J101-V and J102-W).

(b) Press the TEST button and observe the lighting of the BASE light in position X, the EMITTER light in position Y and the GO light.

(c) RESET the module to READY and change the transistor leads to; emitter in position X, collector in position Y, and base in position 2.

(d) Press the TEST button and observe the lighting of the BASE light in position Z, the EMITTER light in position X and the GO light.

(e) RESET the module to READY and remove the test transistor. Switch to DIODE. Place a known good general purpose silicon siode such as the 1N538 or equivalent in parallel with the 9100 ohm test resistor in the front panel DIODE test clips (J101-V and J102-W) with the diode cathode in J101-V and the anode in J102-W.

(f) Press the TEST button and observe the lighting of the CATHODE light in position V, the ANODE light in position W and the GO light.

(g) RESET the module to READY and reverse the position of the diode leads in the DIODE test clips. Do not move the test resistor.

(h) Press the TEST button and observe the lighting of the CATHODE light in position W, the ANODE light in position V and the GO light.

(i) RESET the module to READY and replace the 9100 ohm test resistor with one of 500 ohms plus or minus 1 percent.

(j) Press the TEST button and observe the lighting of the SHORT OR LOW IMP (low impedance) light.

(k) RESET the module to READY and place the IN-CRT IMP (in-circuit impedance) switch in the LOW position and switch to IN-CIRCUIT test.

(l) Press the TEST button and observe the lighting of the CATHODE light in position W, the ANODE light in position V and the GO light.

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(m) RESET the module to READY and replace the test resistor with one of 250 ohms plus or minus 1 percent.

(n) Press the TEST button and observe the lighting of the SHORT OR LOW IMP (low impedance) light.

(o) RESET the module to READY and remove the test diode but leave the 250 ohm test resistor in the DIODE test clips. Place the known good general purpose germanium PNP transistor used in (a) above into the TRANSISTOR test socket with the emitter in position X, the base in position Y, and the collector in position Z. Switch back to TRANSISTOR.

(p) Press the TEST button and observe the lighting of the BASE light in position Y, the EMITTER light in either position X or Z, and the GO light.

(q) RESET the module to READY and replace the 250 ohm test resistor with one of 47 ohms plus or minus 5 percent.

(r) Press the TEST button and observe the lighting of the SHORT OR LOW IMP (low impedance) light.

(s) RESET the module to READY and remove the 47 ohm test resistor and any one lead of the test transistor.

(t) Press the TEST button and observe the lighting of the OPEN light. Disregard the lighting of all three BASE lights.

(u) RESET the module to READY and remove the test transistor.

4.9.2.2.3 Emitter finding.

(a) Switch to NPN, OUT-OF-CIRCUIT test, place the IN-CKT IMP (in-circuit impedance) switch in the HIGH position, and place a known good general purpose silicon NPN transistor such as a 2N338 or equivalent into the front panel TRANSISTOR test socket (J103) with the base in position X, the collector in position Y and the emitter in position Z.

(b) Press the TEST button and observe the lighting of the BASE light in position X, the emitter light in position Z and the GO light.

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(c) RESET the module to READY and place a 500 ohm plus or minus 1 percent test resistor across the front panel DIODE test clips (J101-V and J102-W). Switch to IN-CIRCUIT test and place the IN-CKT IMP (in-circuit impedance) switch in the LOW position. Relocate the leads of the test transistor to have the collector in position X, the emitter in position Y, and the base in position Z.

(d) Press the TEST button and observe the lighting of the BASE light in position Z, the EMITTER light in either position X or Y and the GO light.

(e) RESET the module to READY and replace the 500 ohm test resistor with one of 47 ohms plus or minus 5 percent in the same location.

(f) Press the TEST button and observe the lighting of the BASE light in position Z and the NO GAIN light.

(g) RESET the module to READY and remove the test resistor and test transistor.

(h) The test routine of 4.9.2.2.2 has exercised the other capabilities of this test circuit. Check the results of emitter finding action during tests of 4.9.2.2.2 (b) and (d), and if correct, emitter finding has been obtained, then this test circuit may be considered acceptable.

4.9.2.2.4 PNP - NPN finding.

(a) Use the following test switch line-up: Switch to PNP: TRANSISTOR; LEADS UNKNOWN; IN-CIRCUIT; IN-CKT IMP (in-circuit impedance) switch in LOW position. Insert a known good general purpose silicon NPN transistor such as a 2N338 or equivalent into the front panel TRANSISTOR test socket (J103) with the base in position X, the emitter in position Y and the collector in position Z. Place a 500 ohm plus or minus 1 percent test resistor across the front panel DIODE test clips (J101-V and J102-W).

(b) Press the TEST button and observe the lighting of the BASE light in position X and the EMITTER light in either position Y or Z and the lighting of the NPN light and GO light.

(c) RESET the module to READY. The test routine of 4.9.2.2.2 has exercised the other capabilities of this test circuit. Check the results of lead finding action of PNP transistors during test 4.9.2.2.2 (b) and (d), and if correct, lead finding has been obtained, then this test circuit may be considered acceptable.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. - Preservation, packaging and packing shall be in accordance with Specification MIL-P-14232 and the applicable packaging data sheet. The levels of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The module is for use in the Transistor Test Set AN/USM-136 which is an instrument used to test transistor and semiconductor devices which fall into the design limits of the instrument.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Applicable packaging data sheet

(c) Selection of applicable level of preservation, packaging and packing

(d) Applicable stock number

(e) Provisions for initial production samples including date of delivery (See 3.1).

6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) is defined as the common type measuring devices which are usually stocked by commercial supply houses ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.3 Initial production sample. - A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

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Automatic Lead Selector - In-Circuit Checker (A100) Module F8625449

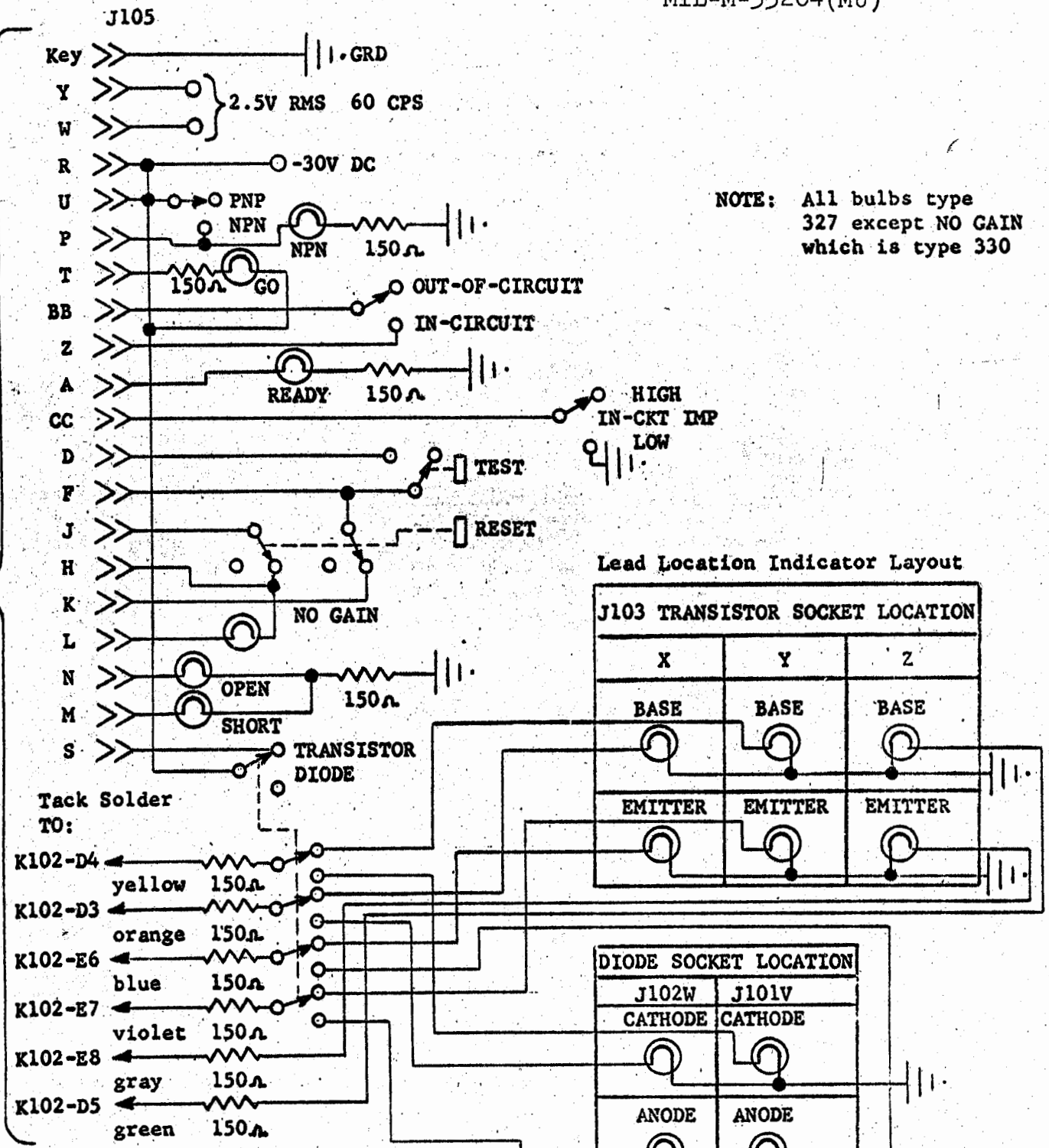


Figure 1

AUTOMATIC LEAD SELECTOR - IN-CIRCUIT CHECKER TEST CIRCUIT

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